**Experiment No.:** 4 Name: Rohan Laharwani PRN: 24070521055

## 1. Experiment Title:

Use function types, scope, and closures; apply try-catch; build a palindrome checker.

## 2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS Javascript

## 3. Experiment Program Code:

***Experiment 4.html***

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Palindrome Checker</title>
    <link rel="stylesheet" href="Experiment 4.css">
</head>

<body>
     <div class="container">
         <div class="card">
             <h2>Palindrome Checker</h2>
             <form id="palindromeForm" onsubmit="event.preventDefault(); checkPalindrome();">
                 <div class="form-group">
                     <label for="word">Enter a String</label>
                     <input type="text" id="word" placeholder="Enter word or text">
                 </div>

                 <input type="button" class="btn" value="Check Palindrome" onclick="checkPalindrome()">
             </form>

             <div id="result"></div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px
solid lightgray; padding-top: 15px;">
             Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Experiment 4.js"></script>
</body>

</html>
```

***Experiment 4.css***

```
body {
    background-color: whitesmoke;
    color: black;
    font-family: sans-serif;
```

```
    margin: 0;
    padding: 0;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}

.container {
    width: 100%;
    max-width: 400px;
    padding: 20px;
}

.card {
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 12px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 20px;
}

hr {
    border: 0;
    height: 1px;
    background-color: lightgray;
    margin: 20px 0;
}

form {
    display: flex;
    flex-direction: column;
    gap: 15px;
}

.form-group {
    display: flex;
    flex-direction: column;
    gap: 5px;
}

label {
    font-weight: bold;
    font-size: 14px;
    color: dimgray;
}

input[type="text"] {
    padding: 10px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 16px;
}

input[type="text"]:focus {
    border-color: blue;
    outline: none;
}

.btn,
input[type="button"] {
    padding: 12px;
```

```
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    transition: background-color 0.2s;
    width: 100%;
}

.btn:hover,
input[type="button"]:hover {
    background-color: blue;
}

#result {
    display: none;
    margin-top: 20px;
}

.result-details p {
    margin: 6px 0;
    font-size: 16px;
}

.status-pass {
    color: green;
}

.status-fail {
    color: red;
}
```

***Experiment 4.js***

```
function reverseString(str) {
    return str.split("").reverse().join("");
}

function message() {
    var msg = "Palindrome Checker";
    function display() {
        return msg;
    }
    return display;
}

function checkPalindrome() {
    try {
        var word = document.getElementById("word").value.trim();
        var resultDiv = document.getElementById("result");

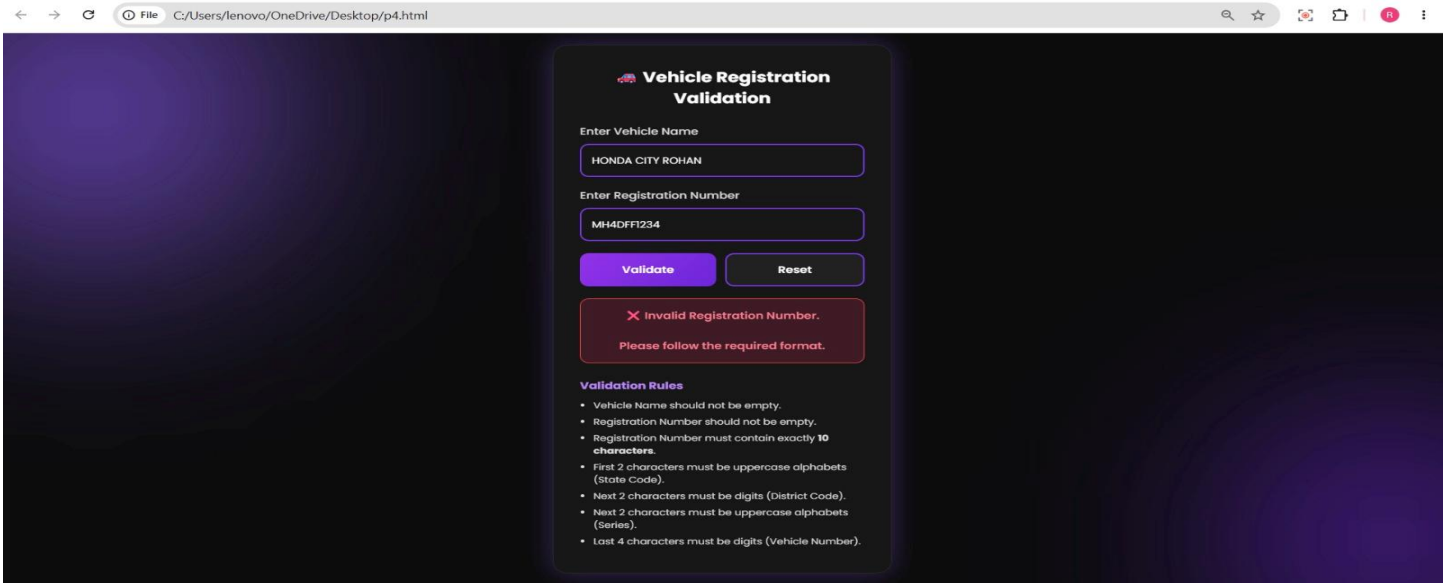
        if (word == "") {
            throw "Please enter a string";
        }
        for (let i = 0; i < word.length; i++) {
            let ch = word[i];
            let code = ch.charCodeAt(0);
            let isAlphaNumeric = (code >= 48 && code <= 57) ||
                (code >= 65 && code <= 90) ||
                (code >= 97 && code <= 122);
            if (!isAlphaNumeric) {
                throw "Only alphabets/digits are allowed!";
            }
        }
        if (word.length < 3) {
            throw "Please enter at least 3 characters";
```

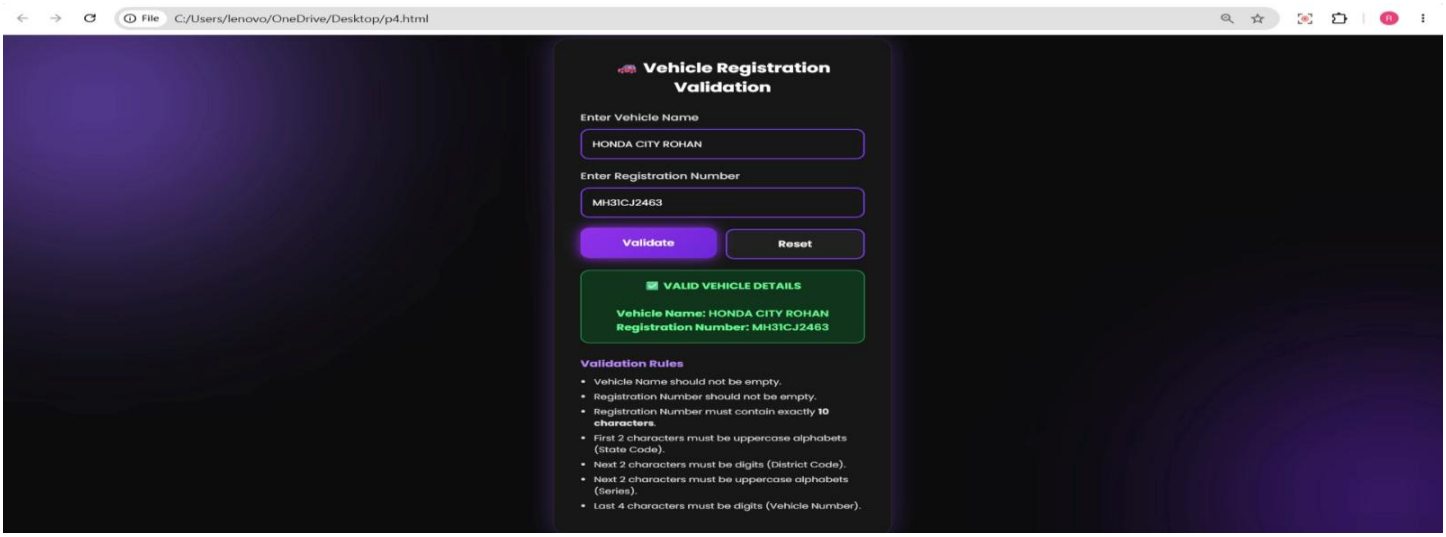
```
        }
        if (word.length > 30) {
            throw "Maximum 30 characters allowed";
        }

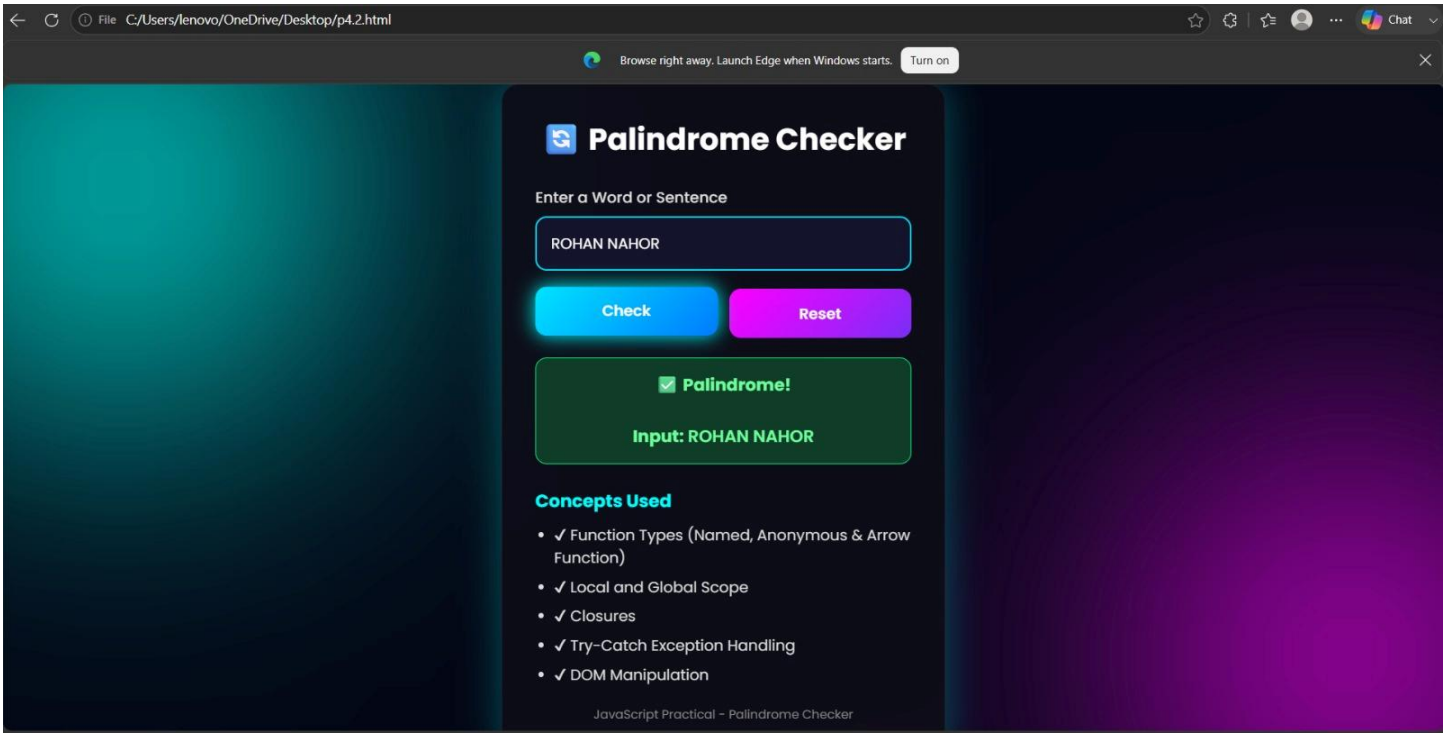
        let input = word.toLowerCase();
        var reverse = reverseString(input);
        var isPalindrome = (input === reverse);

        resultDiv.style.display = "block";
        resultDiv.className = isPalindrome ? "status-pass" : "status-fail";
        resultDiv.innerHTML = `
            <hr>
            <div class="result-details">
                <p><strong>Word:</strong> ${word}</p>
                <p><strong>Reversed:</strong> ${reverse}</p>
                <p><strong>Result:</strong> ${isPalindrome ? "PALINDROME" : "NOT A PALINDROME"}</p>
            </div>
        `;
    }
    catch (error) {
        alert(error);
    }
}
```

## 4. Output:


File C:/Users/lenovo/OneDrive/Desktop/p4.html
Vehicle Registration Validation
Enter Vehicle Name
HONDA CITY ROHAN
Enter Registration Number
MH4DFF1234
Validate
Reset
Invalid Registration Number.
Please follow the required format.
Validation Rules
Vehicle Name should not be empty.
Registration Number should not be empty.
Registration Number must contain exactly 10 characters.
First 2 characters must be uppercase alphabets (State Code).
Next 2 characters must be digits (District Code).
Next 2 characters must be uppercase alphabets (Series).
Last 4 characters must be digits (Vehicle Number).


File C:/Users/lenovo/OneDrive/Desktop/p4.html
Vehicle Registration Validation
Enter Vehicle Name
HONDA CITY ROHAN
Enter Registration Number
MH31CJ2463
Validate
Reset
VALID VEHICLE DETAILS
Vehicle Name: HONDA CITY ROHAN
Registration Number: MH31CJ2463
Validation Rules
Vehicle Name should not be empty.
Registration Number should not be empty.
Registration Number must contain exactly 10 characters.
First 2 characters must be uppercase alphabets (State Code).
Next 2 characters must be digits (District Code).
Next 2 characters must be uppercase alphabets (Series).
Last 4 characters must be digits (Vehicle Number).


File C:/Users/lenovo/OneDrive/Desktop/p4.2.html
Chat
Browse right away. Launch Edge when Windows starts.
Turn on
Palindrome Checker
Enter a Word or Sentence
ROHAN NAHOR
Check
Reset
Palindrome!
Input: ROHAN NAHOR
Concepts Used
✓ Function Types (Named, Anonymous & Arrow Function)
✓ Local and Global Scope
✓ Closures
✓ Try-Catch Exception Handling
✓ DOM Manipulation
JavaScript Practical - Palindrome Checker

## 5. Result/Conclusion:

Function types, variable scope and closures were demonstrated using a nested function (message/display) that returns an inner function retaining access to its outer variable, illustrating closure behaviour. A palindrome checker was built using a named function to reverse the input string, with try-catch blocks used to handle invalid input such as empty strings, non-alphanumeric characters, and strings outside the allowed length range, raising and catching custom error messages via throw. This experiment reinforced the concepts of function scope, closures and exception handling in JavaScript.